

0.1 Determining the conditional mode

The iteratively reweighted least squares (IRLS) algorithm is an efficient method of determining the maximum likelihood estimates of the coefficients in a generalized linear model. We extend it to a *penalized iteratively reweighted least squares* (PIRLS) algorithm for determining the conditional mode, $\tilde{\mathbf{u}}_{\beta, \theta}$ (eq. ??). This algorithm has the form

1. Given parameter values, β and θ , and starting estimates, $\mathbf{u}_0$, of the spherical random effects, evaluate the linear predictor, η , the corresponding conditional mean, $\mu_{\mathcal{Y}|\mathcal{U}=\mathbf{u}}$, and the conditional variance. Establish the weights as the inverse of the variance. We write these weights in the form of a diagonal weight matrix, $\mathbf{W}$, although they are stored and manipulated as a vector.

2. Let

$$Q(\mathbf{u}) = \left\| \mathbf{W}^{1/2} (\mathbf{y} - \mu_{\mathcal{Y}|\mathcal{U}=\mathbf{u}}) \right\|^2 + \|\mathbf{u}\|^2. \quad (1)$$

Solve the penalized, weighted, nonlinear least squares problem

$$\arg \min_{\mathbf{u}} (Q(\mathbf{u})). \quad (2)$$

3. Update the weights, $\mathbf{W}$, and check for convergence. If not converged, go to step 2.

We use a Gauss-Newton algorithm with an orthogonality convergence criterion (?, §2.2.3) to solve the penalized, weighted, nonlinear least squares problem in step 2. At each step of the algorithm the increment to the conditional modes $\delta^{(i)}$ ($\mathbf{u}^{(i+1)} = \mathbf{u}^{(i)} + \delta^{(i)}$) satisfies

$$\mathbf{P} \left(\Lambda_{\theta}^{\top} \mathbf{Z}^{\top} \mathbf{M}^{(i)} \mathbf{W} \mathbf{M}^{(i)} \mathbf{Z} \Lambda_{\theta} + \mathbf{I}_q \right) \mathbf{P}^{\top} \delta^{(i)} = \Lambda_{\theta}^{\top} \mathbf{Z}^{\top} \mathbf{M}^{(i)} \mathbf{W} (\mathbf{y} - \mu^{(i)}) - \mathbf{u}^{(i)} \quad (3)$$

where $\mathbf{u}^{(i)}$ is the current value of $\mathbf{u}$, $\mu^{(i)}$ is the corresponding conditional mean of $\mathcal{Y}|\mathcal{U} = \mathbf{u}^{(i)}$ and $\mathbf{M}^{(i)}$ is the Jacobian matrix of the vector-valued inverse link, evaluated at $\mu^{(i)}$. That is

$$\mathbf{M}^{(i)} = \left. \frac{d\mu}{d\eta^{\top}} \right|_{\eta^{(i)}}, \quad (4)$$

which will be a diagonal matrix so, as for the weights, we store and manipulate the Jacobian as a vector.

We solve (3) using the sparse Cholesky factor. At convergence, the factor, $\mathbf{L}_{\beta, \theta}$, satisfies

$$\mathbf{L}_{\beta, \theta} \mathbf{L}_{\beta, \theta}^{\top} = \mathbf{P} \left(\Lambda_{\theta}^{\top} \mathbf{Z}^{\top} \mathbf{M} \mathbf{W} \mathbf{M} \mathbf{Z} \Lambda_{\theta} + \mathbf{I}_q \right) \mathbf{P}^{\top}, \quad (5)$$

where $\mathbf{P}$ denotes a fill-reducing permutation that is computed based on the sparsity pattern of $\mathbf{Z} \Lambda_{\theta}$ prior to the optimization process (see ?).

When solving for the minimum of $Q(\mathbf{u})$, the proposed update for $\mathbf{u}$ may increase rather than decrease the value of the objective function $Q(\mathbf{u})$, if the step size $\delta^{(i)}$ estimated by the Gauss-Newton step is too large. Let $Q(\mathbf{u}^{(i)})$ represent the value of the objective function $Q(\mathbf{u})$ at the current iterate $\mathbf{u}_i$; similarly, let $Q(\mathbf{u}^{(i-1)})$ represent the value of the objective function of the previous iteration.

If $Q(\mathbf{u}^{(i+1)}) > Q(\mathbf{u}^{(i)})$, we proceed with a *step-halving* procedure. That is, we successively consider scaled updates of the form

$$\mathbf{u}^{(i,j)} = \mathbf{u}^{(i)} + \frac{\delta^{(i)}}{2^j}, \quad j = 1, 2, \dots, \quad (6)$$

computing $Q(\mathbf{u}^{(i,j)})$ at each step. The step-halving procedure continues until a value j^* of j is found such that $Q(\mathbf{u}^{(i,j^*)}) < Q(\mathbf{u}^{(i)})$, after which the iterate is then updated by setting $\mathbf{u}^{(i+1)} = \mathbf{u}^{(i,j^*)}$.

If no such j is found after a predetermined number of halvings, then the step-halving procedure has failed and the algorithm terminates.

As we show in the next section, the matrix $(\mathbf{L}_{\beta,\theta} \mathbf{L}_{\beta,\theta}^\top)^{-1}$ is a Laplace approximation of the covariance matrix for the spherical random effects, conditional on the observed data. This fact is useful for constructing a nonlinear objective function for finding the approximate maximum likelihood estimates of θ and β .